Optimizing UPI AutoPay: Reducing Payment Failures During Peak Hours

28%

of mandate debits fail;

55%

occur in 10 AM-1 PM peak window

Simulated data analysis for smarter debit recovery

Failed Debits Quietly Become Lost Customers

Every failed auto-pay isn’t just a payment issue—it’s revenue leakage and customer churn.

Scheduled GTV	First Attempts Collected	Naive Retries Recovered	Lost Outright
₹1.44 crore	→ 75.8% collected	→ 28.7L recovered	→ ₹6.1L (4.3%) lost

Revenue leakage across the debit journey



After two failed’s,

61.6%

of users churn.

Recovered users show

near zero

churn.

Regulation Made Timing a Product Constraint



RBI/NPCI rules define the debit lifecycle.
Smart timing within these windows protects both recovery and compliance.

PRE-DEBIT NOTICE
T-24h
Customer informed before debit attempt

DEBIT
NOW
Original debit attempt window

10 AM – 1 PM
PEAK WINDOW

RETRY WINDOW
T+24h
Retry permitted within window

Operating outside these timings can lead to declines, penalties, and compliance risk.

Verify against current RBI/NPCI circulars

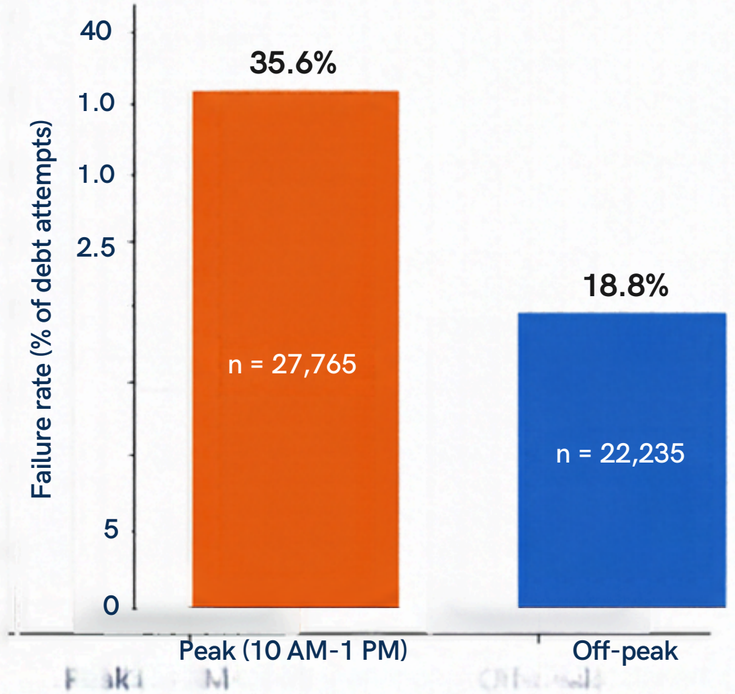
FUNNEL DEEP DIVE

Peak hours drive nearly 2x more failures

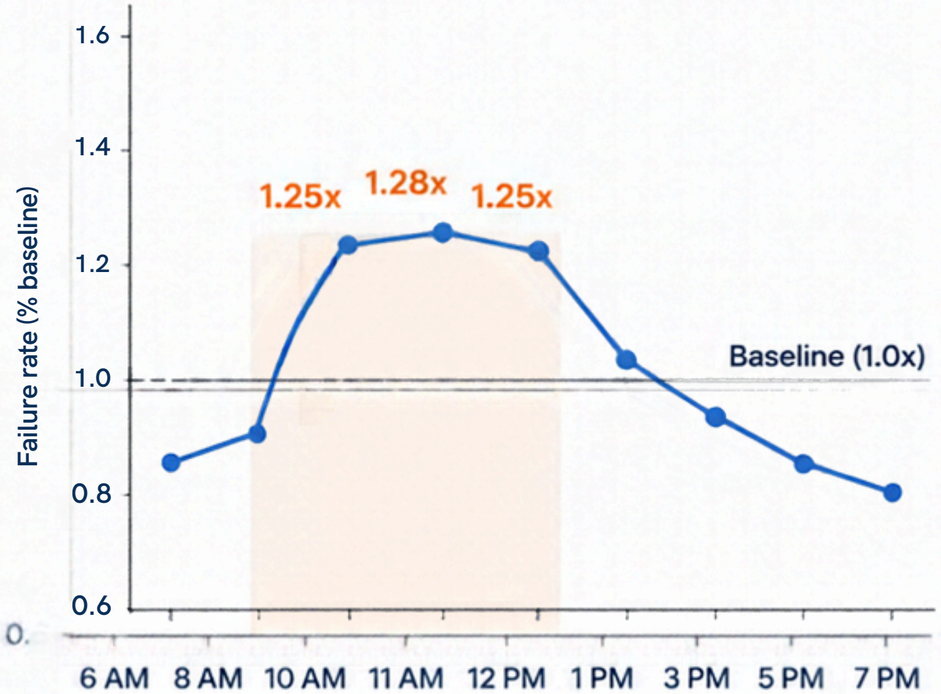
Congestion at NPCI peak hours leads to higher mandate debit failures and involuntary churn.



Mandate debits fail far more often at NPCI peak hours (simulated data)



Hourly failure rate spikes during NPCI peak window



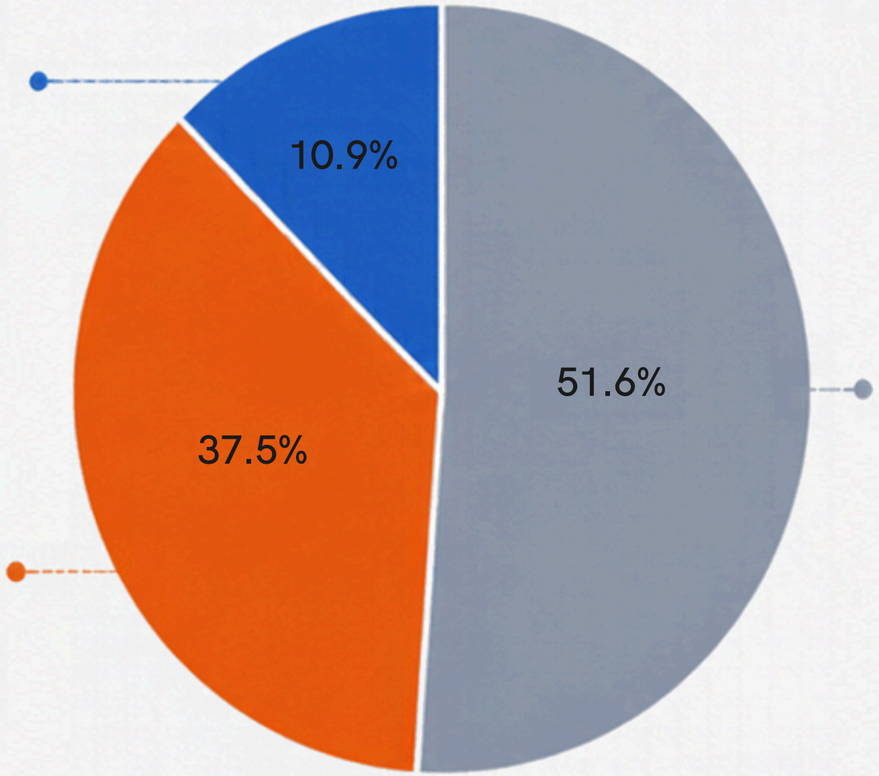
Failures at peak hours are ~2x higher than off-peak, driving revenue leakage and involuntary churn.

Failure mix

System errors drive more than half of mandate debit failures.

Process errors
10.9%
Notification and process gaps

User errors
37.5%
Insufficient funds and timeouts

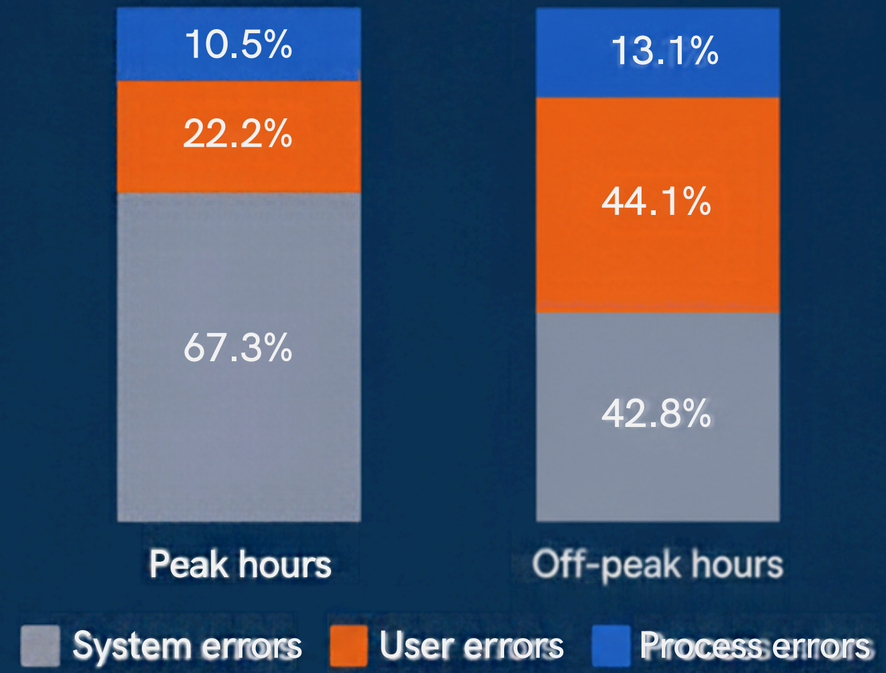


System errors
51.6%
Peak-hour congestion

52% system **38% user** **11% process**

Failures by time of day

System errors dominate during peak hours.



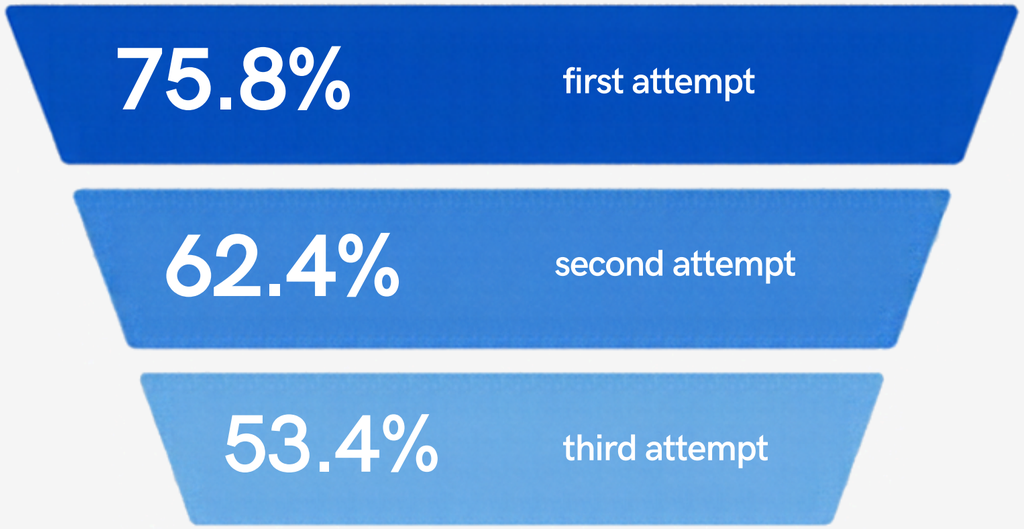


Peak-hour congestion creates system failures beyond customer control.

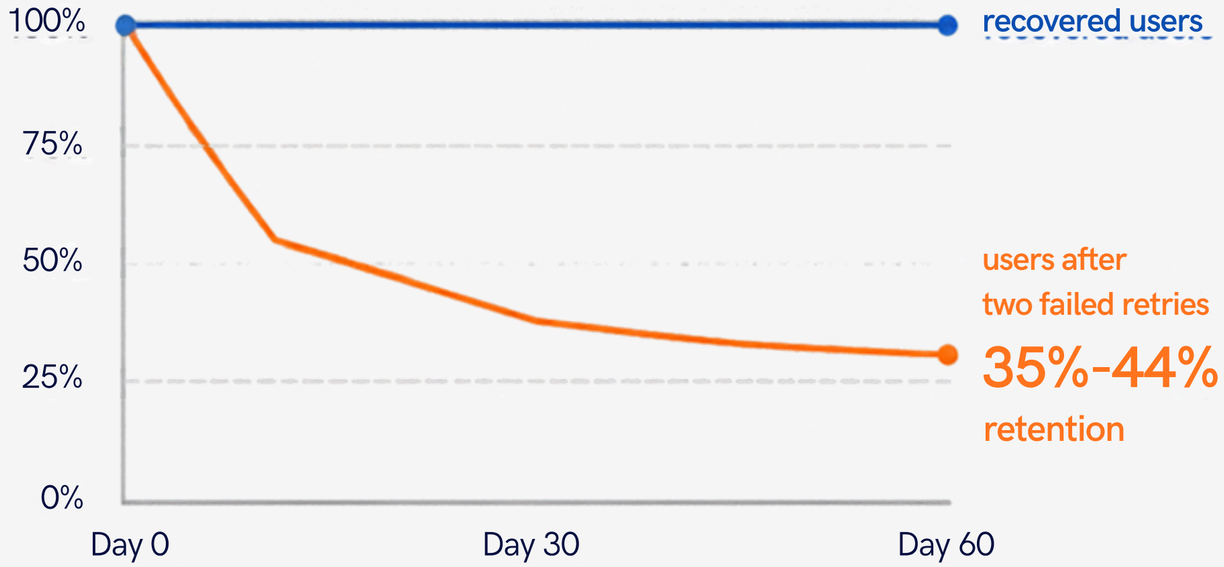
Smart retries during low-traffic windows can recover more revenue.

Naive Retries Decay and Churn Follows

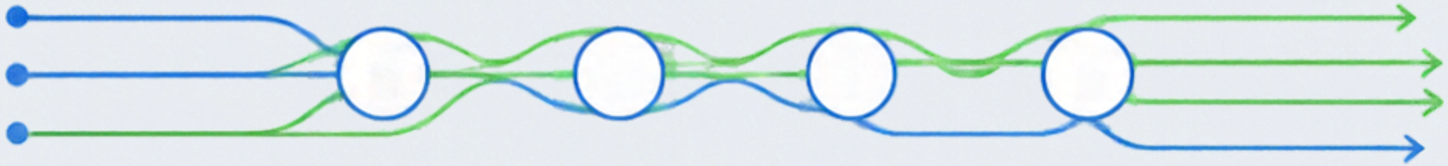
SUCCESS RATE DECLINES WITH EACH RETRY



RETENTION DROPS AFTER FAILED RETRIES



Smart retries in off-peak windows recover more mandates and protect long-term retention.



RICE Prioritization Engine

Data-driven ranking to focus MVP on the highest-impact wins

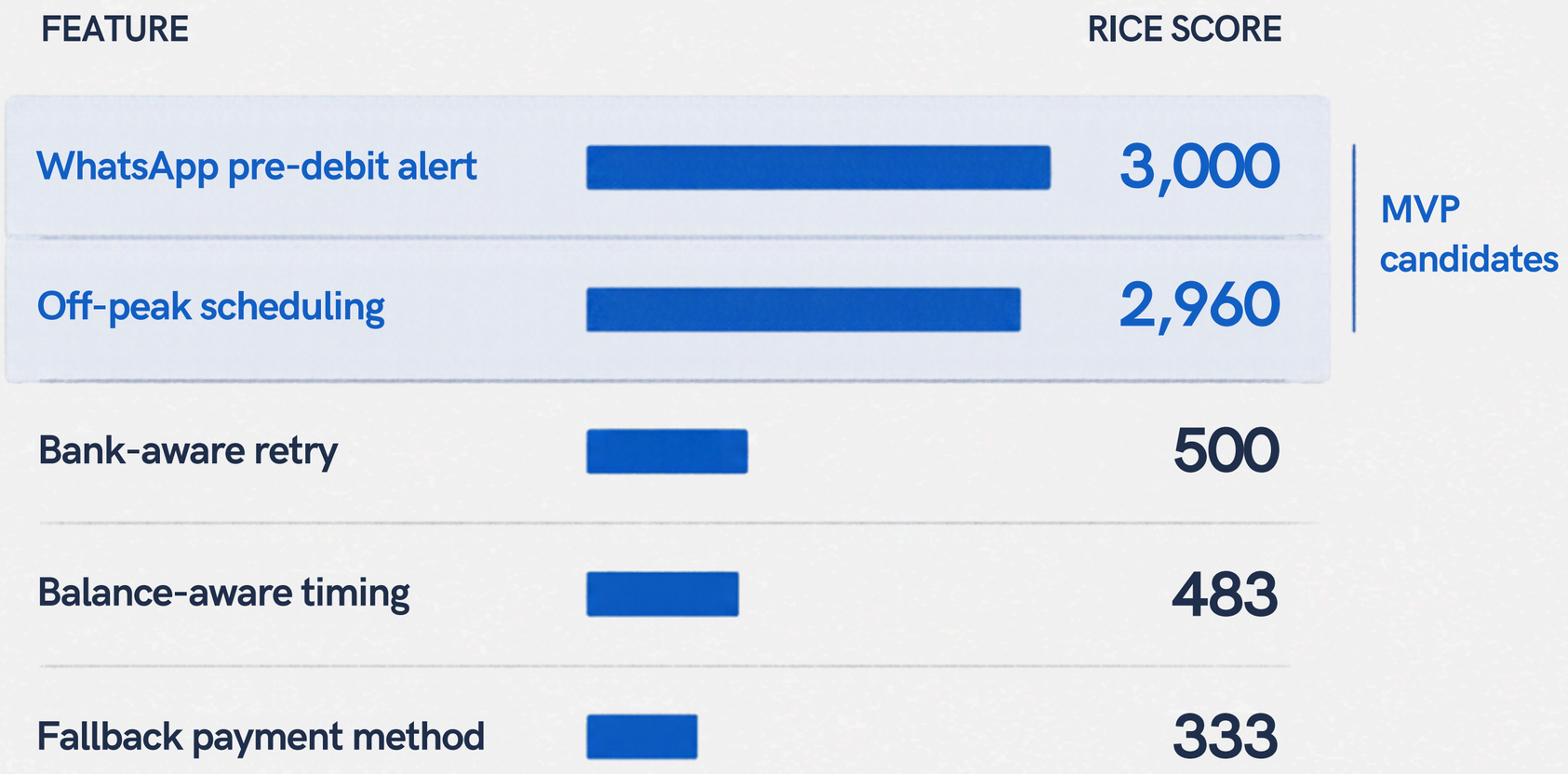
What is RICE?

RICE Score = Reach × Impact × Confidence + Effort

Higher scores mean more value delivered for the effort invested.

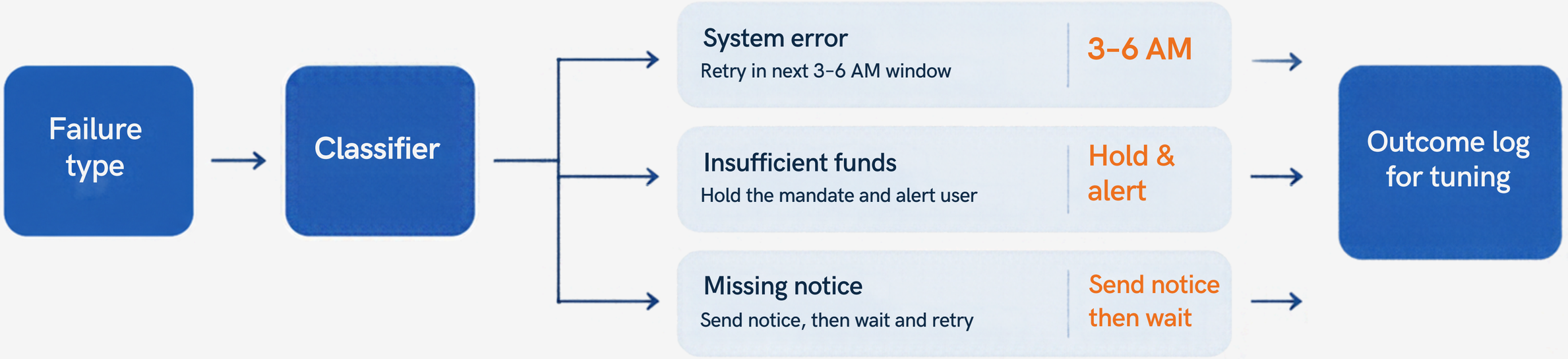
Scoring assumptions

Based on simulated mandate-debit failure data, customer value, and implementation effort.



The Smart-Retry Engine

Intelligent retries. Fewer failures. Higher success.



Real-time WhatsApp alert reminds users to

maintain balance

before debit

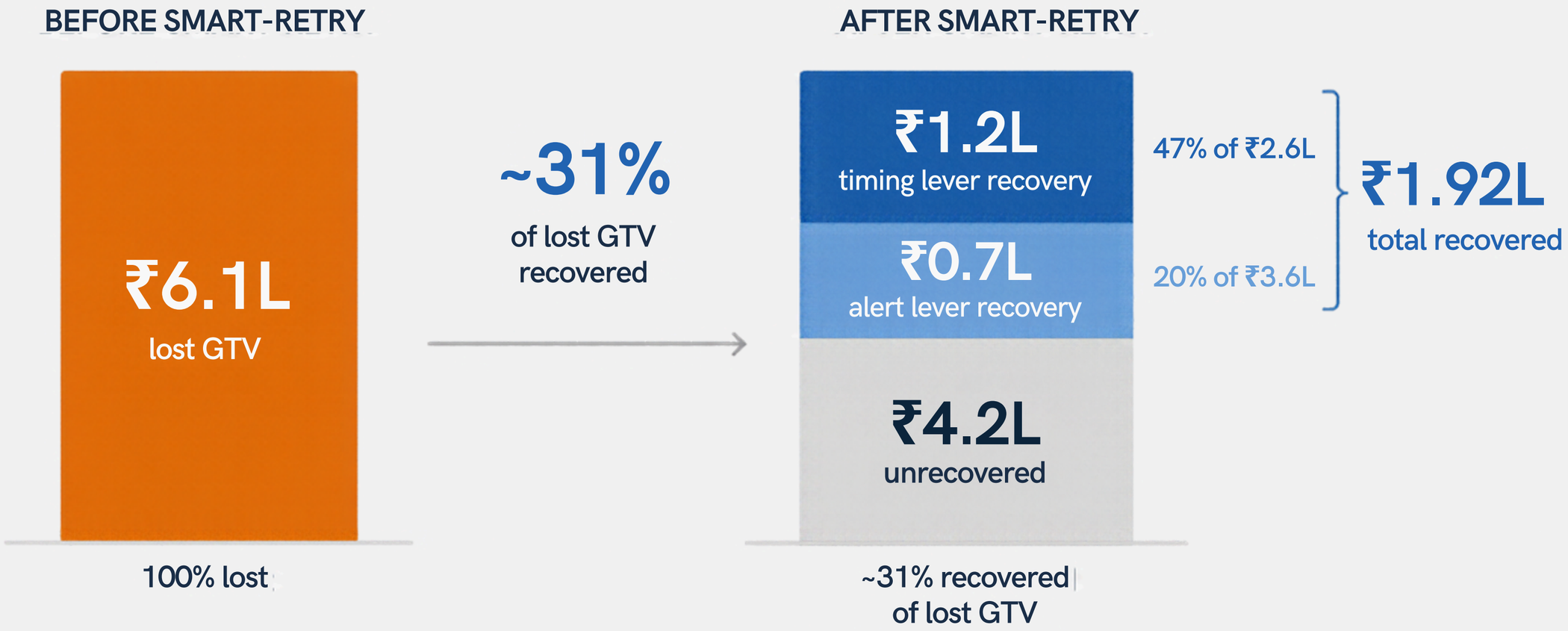


GTV Recovery

Smart-Retry Engine recovers ₹1.9L of lost GTV

₹1.9L

projected on ₹6.1L lost



Note: Conservative estimate. excludes benefits from faster collections and future churn losses.

North Star

Success, Safety, Scale

RTSR
Recurring Transaction Success Rate

Complaint Rate
Keep customer friction low

Chargebacks
Protect against financial risk

Notification Opt-outs
Maintain consent and trust

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01 10% Pilot
10% pilot with compliance sign-off

→

02 Tune & Expand
Bank-aware tuning and feature phase 2



→

03 Scale & Measure
Full rollout with A/B testing and measurement